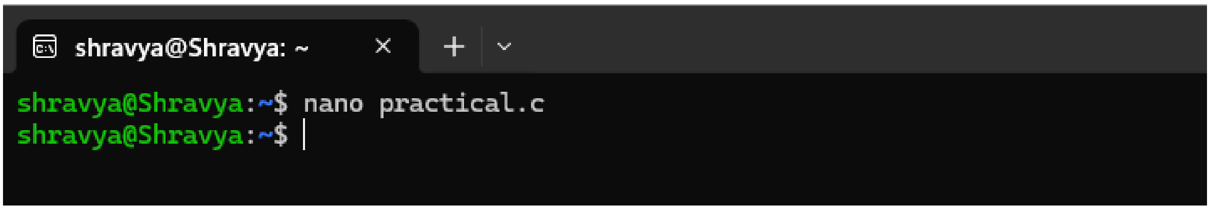
OSSP-PRACTICAL-1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

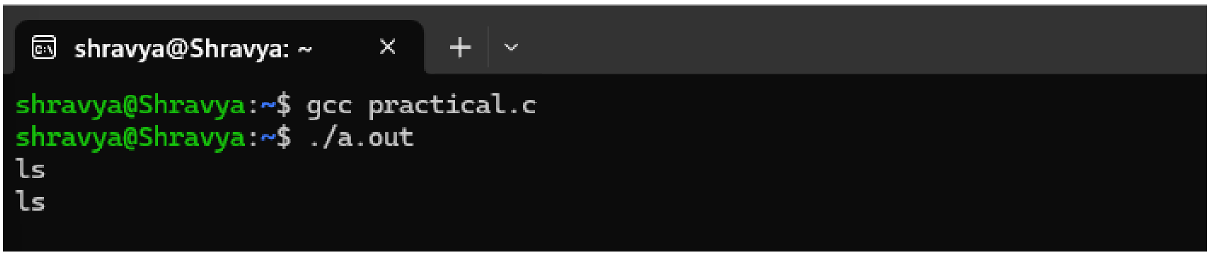
Accept a Linux command as input.

- Created a file named practical.c using



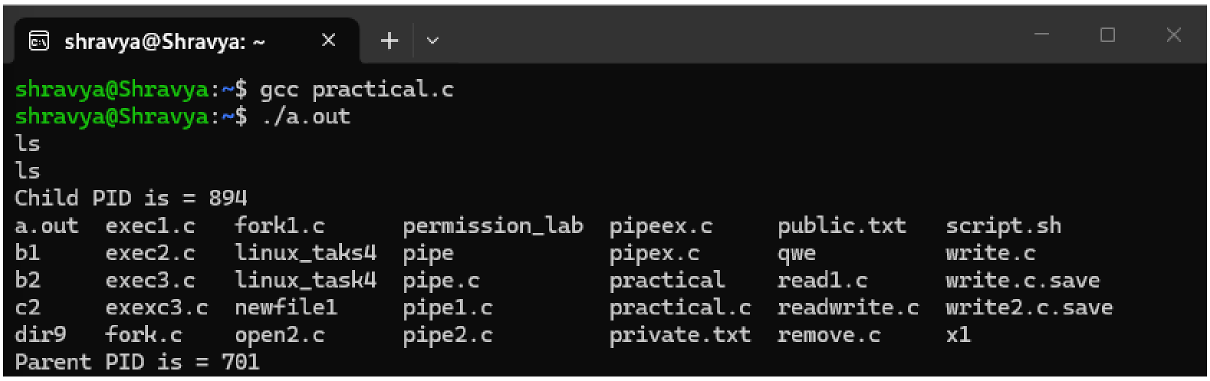
Execute the command in the child process using an appropriate exec() system call.

- Compiled and ran the program where the command “ls” is entered.



Allow the parent process to wait for the child using wait ().

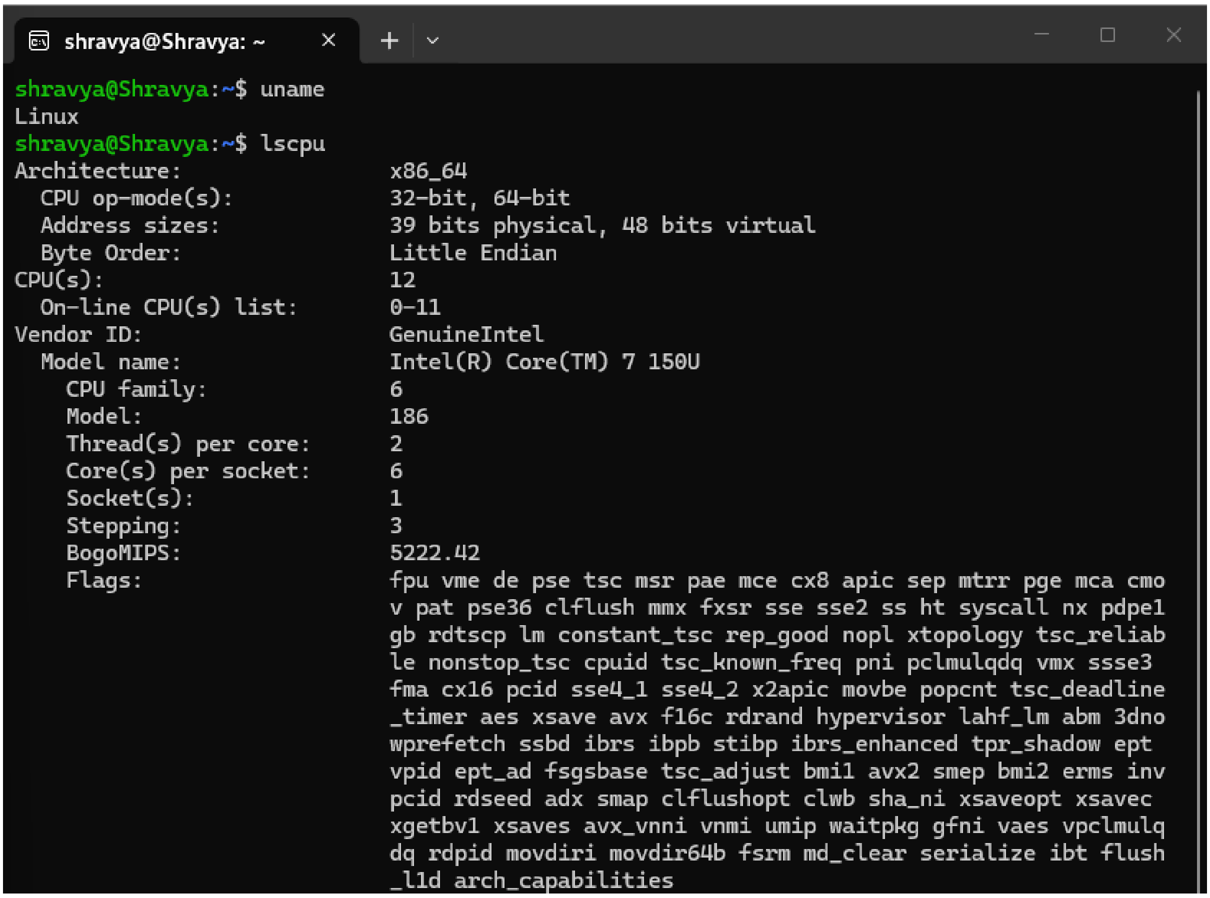
- The Child PID has been returned followed by its command execution while the parent process waited till the child process completes.



```
shravya@Shravya: ~
shravya@Shravya:~$ gcc practical.c
shravya@Shravya:~$ ./a.out
ls
ls
Child PID is = 894
a.out  exec1.c   fork1.c      permission_lab  pipeex.c     public.txt    script.sh
b1     exec2.c   linux_taks4  pipe            pipex.c      qwe           write.c
b2     exec3.c   linux_task4  pipe.c          practical    read1.c       write.c.save
c2     exexc3.c  newfile1     pipe1.c         practical.c  readwrite.c   write2.c.save
dir9   fork.c    open2.c      pipe2.c         private.txt  remove.c      x1
Parent PID is = 701
```

Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigated the relationship between hardware resources and operating system services



```
shravya@Shravya: ~
shravya@Shravya:~$ uname
Linux
shravya@Shravya:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   12
  On-line CPU(s) list:    0-11
Vendor ID:                GenuineIntel
  Model name:             Intel(R) Core(TM) 7 150U
    CPU family:           6
    Model:                186
    Thread(s) per core:   2
    Core(s) per socket:   6
    Socket(s):            1
    Stepping:             3
    BogoMIPS:             5222.42
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmo
                          v pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1
                          gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliab
                          le nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3
                          fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline
                          _timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dno
                          wprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept
                          vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms inv
                          pcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec
                          xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulq
                          dq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush
                          _l1d arch_capabilities
```

vida@LAPTOP-4B7UUAUH: ~

```
                             arch_capabilities
Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       288 KiB (6 instances)
  L1i:                       192 KiB (6 instances)
  L2:                        7.5 MiB (6 instances)
  L3:                        12 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-11
Vulnerabilities:
  Gather data sampling:      Not affected
  Ghostwrite:                Not affected
  Indirect target selection: Not affected
  Itlb multihit:             Not affected
  L1tf:                      Not affected
  Mds:                       Not affected
  Meltdown:                  Not affected
  Mmio stale data:           Not affected
  Old microcode:             Not affected
  Reg file data sampling:    Mitigation; Clear Register File
  Retbleed:                  Mitigation; Enhanced IBRS
  Spec rstack overflow:      Not affected
  Spec store bypass:         Mitigation; Speculative Store Bypas
                             s disabled via prctl
  Spectre v1:                Mitigation; usercopy/swapgs barrier
                             s and __user pointer sanitization
  Spectre v2:                Mitigation; Enhanced / Automatic IB
                             RS; IBPB conditional; PBRSB-eIBRS S
                             W sequence; BHI BHI_DIS_S
  Srbds:                     Not affected
  Tsa:                       Not affected
  Tsx async abort:           Not affected
  Vmscape:                   Not affected
```

```
vida@LAPTOP-4B7UUAUH:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                                /
vida@LAPTOP-4B7UUAUH:~$ ps
    PID TTY          TIME CMD
    317 pts/0    00:00:00 bash
    481 pts/0    00:00:00 ps
vida@LAPTOP-4B7UUAUH:~$ top
top - 17:01:29 up  1:14,  1 user,  load average: 0.04, 0.05, 0.01
Tasks:  25 total,   1 running,  24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7788.6 total,   7218.7 free,    555.5 used,    165.5 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7233.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24320  15364  11608 S   0.0   0.2   0:00.61 systemd
      2 root      20   0    3180   2208   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
      7 root      20   0    3180   2040   1940 S   0.0   0.0   0:00.00 init
     47 root      19  -1   42200  16324  15052 S   0.0   0.2   0:00.14 systemd-journal
     79 systemd+  20   0   22540  14600  11980 S   0.0   0.2   0:00.06 systemd-resolve
     85 root      20   0   35428  12388   9228 S   0.0   0.2   0:00.22 systemd-udevd
    122 root      20   0    2888   1948   1820 S   0.0   0.0   0:00.02 chronyd-starter
    131 root      20   0    4472   3164   2916 S   0.0   0.0   0:00.00 cron
    142 message+  20   0    8816   5348   4676 S   0.0   0.1   0:00.04 dbus-daemon
    153 root      20   0   44092  29700  14560 S   0.0   0.4   0:00.14 networkd-dispat
    157 root      20   0   18436   9444   8196 S   0.0   0.1   0:00.06 systemd-logind
    160 root      20   0 1792300  14184  11512 S   0.0   0.2   0:00.07 wsl-pro-service
    166 syslog    20   0  220548   5596   4660 S   0.0   0.1   0:00.04 rsyslogd
    213 _chrony   20   0   20420   6492   5716 S   0.0   0.1   0:00.04 chronyd
    222 _chrony   20   0   12092   2456   1816 S   0.0   0.0   0:00.00 chronyd
    228 root      20   0  123456  32720  18108 S   0.0   0.4   0:00.10 unattended-upgr
```

1. uname

It displays basic information about the Linux operating system and kernel.

2. lscpu

It displays CPU information such as the number of CPUs, cores, architecture, and processor details.

3. lsblk

It displays the storage devices and partitions available in the system.

4. ps

It shows the processes currently running in the system.

5. top

It shows running processes along with their CPU and memory usage in real time.

Hardware and the Operating System :

The operating system acts as a link between applications and computer hardware.

- **CPU:** The OS schedules processes and assigns them CPU time.
- **Memory:** The OS allocates RAM to processes and keeps their memory spaces separate.
- **Storage:** The OS manages storage devices and organizes data into files and directories.
- **I/O Devices:** The OS uses device drivers to communicate with devices such as keyboards, displays, and network cards.